HUSTLEHUB

RELEASE REPORT | VERSION 1.0.0

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Module	OPSC6312 – Open Source Coding (Advanced)	**Build**	Build 10
Assessment	Part 3 – Final PoE	**Date**	November 2026

Hustle Smart, Grow Fast A freelance business manager built for South African freelancers.

1. Release Overview

HustleHub v1.0 updates the original prototype into a more complete freelance management app. This release focuses on reliable data, secure access, practical project and invoice tools, and a smoother experience for new users. It adds features suited to South African freelancers, including ZAR/USD conversion, WhatsApp invoice sharing and offline-first data handling.

Main release goal: Remove prototype test data, improve stability and ensure information reflects real user activity rather than hardcoded values.

2. Main Improvements

Account security and profile

- Separate registration and login screens, with Google Sign-In.
- Encrypted local preference storage and secure logout.
- Profile picture uploads using Firebase Storage.

Data, storage and offline support

- Room database stores app data locally, while Firestore supports cloud storage and syncing.
- WorkManager syncs local changes when an internet connection is available.
- Database migrations and safer task-data parsing help prevent crashes during updates.

Projects, clients and task management

- Improved Kanban board with five stages: Proposed, In Progress, Done, Invoiced and Paid.
- Project descriptions, deadlines, task checklists and overdue indicators.
- Client phone numbers with a call option, plus add, edit and delete actions.

Dashboard, settings and accessibility

- Dashboard figures are calculated from saved data; new accounts start with zero values.
- Settings include hourly rate, currency, default deadlines, notifications and account options.
- English, Afrikaans and isiZulu language options are available.

3. Key Features

Time tracking and invoicing: A timer continues while the app is in the background. Tracked time can calculate billable amounts, and invoices can be generated as PDFs.

WhatsApp invoice sharing: Invoices can be shared through WhatsApp with a prepared message for the client.

Expenses and profit: Users can record business expenses and compare them with invoice income to estimate profit.

Currency conversion: USD project amounts can be converted to ZAR using ExchangeRate-API.

Notifications: Firebase Cloud Messaging and scheduled deadline checks support reminders.

XP, levels and badges: Task completion, on-time work and paid invoices contribute to XP, levels, streaks and achievement badges.

Load-shedding resilience: Local data remains available offline and can sync when connectivity returns.

4. Technical Implementation

Area	Technology / implementation
Authentication	Firebase Authentication, Google Sign-In and EncryptedSharedPreferences.
Local database	Room database with entities, migrations and local data access.
Cloud services	Cloud Firestore and Firebase Storage.
Background work	WorkManager for syncing and deadline checks; foreground service for the timer.
Notifications	Firebase Cloud Messaging and Android notification permissions.
External API	ExchangeRate-API accessed through Retrofit.
Build and version control	Gradle, GitHub and GitHub Actions automated build workflow.

5. Stability and User Experience

- Removed seeded sample data so each new user starts with a clean account.
- Replaced hardcoded dashboard figures with values calculated from the database.
- Improved Kanban spacing and added the missing Paid stage.
- Added safer handling for missing project IDs and invalid task data.
- Added input limits and feedback for task creation and missing client phone numbers.

6. Testing and Release Evidence

The final app was installed and tested on a physical Samsung A32 device rather than only on an emulator. Screenshots document the main screens, including login, registration, dashboard, Kanban board, clients, project details, timer, invoices, badges, settings, language selection, profile and dark theme.

A signed release APK and Android App Bundle were generated. The APK is recorded at **app/build/outputs/apk/release/app-release.apk**. The project also includes a GitHub Actions workflow that builds the app and runs unit tests.

7. AI Tools Usage

Meta AI and GitHub Copilot were used to assist with code scaffolding, debugging, UI ideas and documentation. Examples include Room database structures and migrations, Firebase Storage upload logic, notification handling and safer task parsing. AI was also used to explore feature ideas for South African freelancers. Generated suggestions were reviewed, adjusted and tested as part of the implementation.

8. Final Deliverables

- HustleHub Android application – v1.0.0 (Build 10).
- Signed release APK and generated AAB.
- Project source code and documentation in the GitHub repository.
- Screenshots of the final app running on a physical Android device.
- GitHub Actions workflow for automated builds and tests.
- Final demonstration video and Figma design reference (where applicable).

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For: OPSC6312 – Final PoE, IIE International
Release: HustleHub v1.0.0 – November 2026